

Physical Human-Robot Interaction Through Hugs with CASTOR Robot

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Abstract. Hugs play an essential role in social bonding between people. This study evaluates the hug interactions with a robot identifying the perception. Four hug release methods in adults were applied, a short-time hug, a long-time hug, a touch-controlled hug, and a pressure-controlled hug. The social robot CASTOR was integrated into this study, a modification was made in its arms to perform the hugging action, and a pressure sensor was in its upper back. 12 adults (5 females and 7 males) participated in the study. Results showed that the perception of friendliness comparing the short-time hug and the pressure-controlled hug had differences ($p = 0.036$), making the pressure-controlled hug more friendly. In the case of natural perception, comparing the short-time hug and the touch-controlled hug, significant differences were obtained too ($p = 0.047$), in which the touch-controlled hug was more natural. This study presents the feasibility of implementing CASTOR in hugging interactions.

Keywords: Physical Human-Robot Interaction, Robotic Hug, Socially Assistive Robotics

1 Introduction

Physical contact is necessary for human beings to maintain psychological, emotional, and bodily well-being. It increases the hormones that produce the sensation of joy and happiness [1]. There is an evolution of physical contact needs, from childhood where complete physical contact is required to the development process to adulthood and elderly, where gestures of affection for emotional support are demanded [2, 3].

Hugging is the main sign of affection. It has different meanings, such as showing friendship, affection, gratitude, and emotional support [4]. Among the associated benefits is reducing stress and tension levels by releasing oxytocin, reducing the risk of dementia by giving tranquility and confidence, reduction of blood pressure by activating Pacini corpuscle receptors in the skin [5]. Additionally, it increases self-esteem and improves mood by generating serotonin and dopamine release, finally, stimulating oxygenation and strengthening the immune system [6].

There are several methods of psychological and emotional therapy. One of these methods is hug therapy, which is defined as the practice of giving or receiving hugs to maintain health or perform specific interventions in health problems [7]. This type of therapy is provided to people with psychological risk since physical contact and especially hugs are fundamental in interpersonal relationships. Therefore, population groups with this limited basic interaction are also suitable to receive hug therapy [8]. Another application is the reduction of insomnia by giving and receiving hugs considerably reduces it. Hugs generate the release of serotonin, which in turn helps to balance melatonin levels [9]. This hormone is associated with sleep and calmness, so one of the therapy methods for some sleep problems is hugging [8].

However, due to the COVID-19 pandemic, this type of psychological and physical support has been limited considerably in society, based on physical distancing and especially in population groups that significantly require these interactions, as in the older population [10]. Such is the need for physical contact in the pandemic that social interactions have changed and adapted to the current context [11]. This social limitation has led to increased levels of depression and anxiety [12]. On the other hand, stress and difficulty sleeping cases have increased [13].

Recently, there has been an increase in research and projects aimed at developing social robots. One application for these social robots has been hugging, where some studies describe the design methods of hug robots [14], the evaluation of the emotional perception after hugging a robot [15], the relationship of human-robot interaction with a stress measure [16] and the relation of the perception with the release method implemented in the hug [17].

Casas et al. [18] developed the CASTOR (CompliAnt Soft Robotics) robot (Center of Biomechatronics, Colombia) from CompliAnt Soft Robotics. CASTOR is a low-cost open-source platform initially created for therapies in children with Autism Spectrum Disorder (ASD) [19]. CASTOR integrates soft actuation and compliant mechanics to provide robustness and safety necessary for physical interaction. This study evaluates the feasibility of implementing CASTOR in hugging interactions. Therefore, the perception in adults of being hugged by the robot in four release methods is analyzed.

2 Related work

There are some studies of robotic hugs in adult population. For instance, *Shiomi et al.* [20] used the social robot Moffuly (JSPS KAKENHI, Japan). It is a robot that resembles a large teddy bear that only moves its arms. The results suggested an increase in people's pro-social behaviors, i.e., hugging increases the voluntary behaviors to help others. Another positive effect of hugging was found by *Block et al.* [15] using the PR2 robot (Willow Garage, USA). The results showed that reciprocated hugs increased the interaction times and encouraged more self-disclosure from the hugged participants than those who did not get reciprocated hugs.

Considering the above results, *Block et al.* [21] evaluated physical variations of warmth and softness in hugs using the PR2 robot again. They measured the pressure exerted by the participants, evaluated the perception through surveys in the physical variations, and changing the duration of the hug i.e., short time (1 second), average time (2.5 seconds), and long time (5 seconds). They determined that the social perception of the robot hug was related to the hug duration between the too-long and the too-short hug. Newly *Block et al.* [17] worked in a hug interaction study, they developed the Huggiebot 2.0 (MPI-IS and ETH, Germany and Switzerland). They evaluated six parameters in the perception of the hug action, one of them was the release methods comparing timed-release hugs and haptic release hugs. The results suggested that the perception of the haptic release method had better results than the timed-release hugs [17].

These studies were from the USA [14], Asia [20], and Europe [15, 21, 17]. Among the robots implemented, there are commercial robots [15, 21], modified stuff teddy bears [20, 14], and social robots developed by investigation groups [17]. This shows the lack of research in robotic hug interactions in developing countries. Moreover, the evaluation methods presented were perceptions surveys in different scenarios, preset duration hugs [21], sensors release methods [17], and the social robot physical characteristics [20, 15].

Considering the above, this work implemented the low-cost Open-Source CASTOR social robot in the social interaction of hugging to evaluate the perception of two time-controlled hugs and two sensors-controlled hugs.

3 Method

As previously stated, this work seeks to evaluate the volunteer’s perceptions in a hug interaction with the socially assistive robot CASTOR. Aiming to achieve this goal, a questionnaire based on previous surveys in HRI research and typical Unified Theory of Acceptance and Use of Technology (UTAUT) questionnaires was used [17]. The study has been divided into two parts. The first focuses on the perception carried out with the volunteers about social robotic, and the second part on the perception of each hug. The first part seeks to evaluate the acceptance of social robotics in 13 questions (i.e., Perceived Safety (PS), Perceived Trust (PT), Attitude Towards Technology (ATT), Anxiety (ANX), Social Influence (SI), Social Presence (SP), Perceived Ease of Use (PEOU) and Perceived Utility (PU)). The second part seeks to evaluate each hug interaction (i.e., Friendliness, Safety, Social Intelligence, Enjoyment, and Natural). This questionnaire is shown in table 1. Each affirmation was scored with a 5-point Likert scale (1: Totally disagree; 2: Disagree; 3: Neutral; 4: Agree; 5: Totally agree).

In the second part, the duration of the touch-controlled hug and the pressure-controlled hug was measured. The touch sensor graphs show two events, one at the beginning and one at the end of the hug, from which the duration was obtained.

Table 1. Questionnaire for each robot hug experience

Evaluation metric	Affirmation
Friendliness	This hug made the robot seem friendly
Safety	This robot behavior seemed safe
Social Intelligence	This hug made the robot seem socially intelligent
Enjoyment	This hug interaction felt enjoyable
Natural	This robot behavior seemed natural

3.1 Participants

A total of 12 healthy subjects (7 males, 5 females, 22.16 ± 2.08 years old, 1.67 ± 0.077 m) performed the study. All subjects were informed about the scope and purpose of the experiment, and written consent was obtained from each of them before the study.

3.2 Equipment

CASTOR robot was modified for the study. In the mechanical part, a change was made on the shoulder motors. This was done to lift the arms for a more extended period. For this purpose, the shoulder piece was modified to add Dynamixel MX-106 motors (Dynamixel, Seoul, Korea), as shown in figure 1A. These motors are characterized by higher RPM and better stall torque than Dynamixel AX-12 motors (Dynamixel, Seoul, Korea).

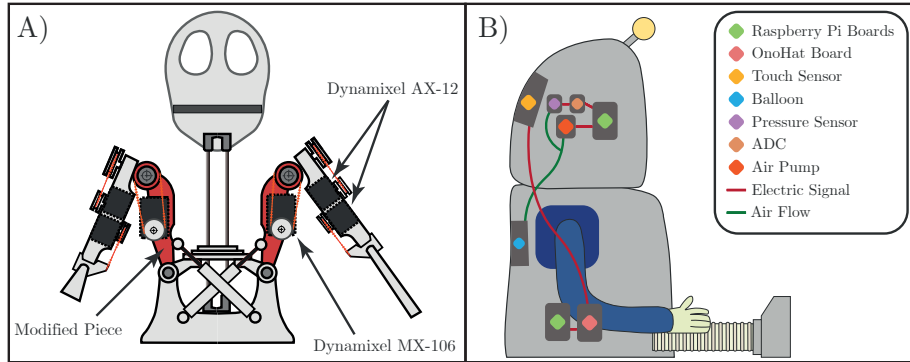


Fig. 1. Mechanic and electronic modifications. A) Mechanic modification of the robot's arms. B) Electronic distribution of the touch and pressure sensors.

On the other hand, a pneumatic system was added. The pneumatic system is composed of a fabric bag and a plastic balloon inside, a pressure sensor ASDX100PAAA5 (Honeywell, North Carolina, USA), and an air pump ROB-10398 (Sparkfun, Colorado, USA). The ADC ADS1115 (Adafruit, New York,

USA) was implemented to read the pressure sensor. Furthermore, the perception module implemented in the CASTOR robot based on touch sensors made of Velostat (Adafruit, New York, USA) was used, specifically the sensor located at the back of the head. The electronic system is shown in figure 1B

Consequently, four movements were established to perform a hug: a neutral position where the robot's arms are close to the thorax, a preparation position to warn the participant to start the hug, the opening of the arms to start the hug, finally, the closing of the arms, as shown in figure 2.

Additionally, three methods were realized to control the opening and closing of the arms. The first method is time-based, where the participant controls the closing of the arms using the touch sensor, and after some time, the arms open. The second method, where the participant controls the opening and closing of the arms using the touch sensor. Finally, the third method where a balloon inflation process is initiated, and the participant controls the opening and closing of the arms by exerting force on the balloon to increase air pressure.

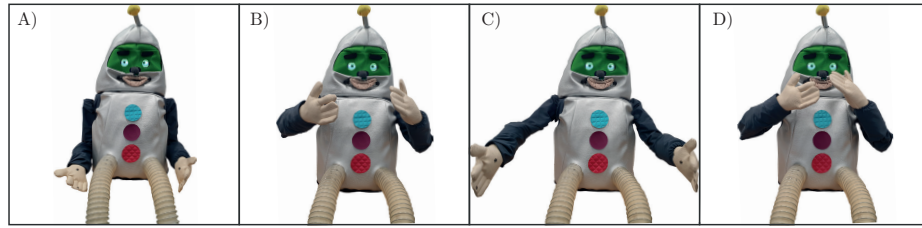


Fig. 2. CASTOR's arms movements to perform a hug. A) Neutral position. B) Preparation position. C) Opening arms. D) Closing arms

Considering that the height of CASTOR is 54 cm, and a suitable height is required according to the participant's height. A setup with three different heights was established, as shown in the figure 3. The height used for each participant corresponds to the next low closest to the participant's height.

For the analysis of the participant's behavior and the robot, the interventions of the 12 subjects in their four interactions with the robot were recorded using two cameras, one located behind the robot and the second located on the interaction side.

3.3 Procedure

Every study spent an estimated time between 15 to 25 minutes. In the beginning, the activity to be performed was explained, the participants read and signed the informed consent form.

Then the participant is conducted inside the room, introduced to the robot, and allowed to interact with it. The participant then takes the social robot

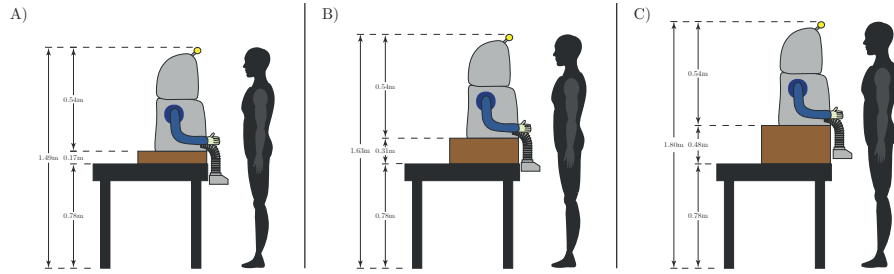


Fig. 3. Heights available for the participants nearest down height. A) The 1.49m height: With the 78cm table and a 17cm box. B) The 1.63m height: With the 78cm table and a 31cm box. C) The 1.80m height: With the 78cm table and a 48cm box.

perception questionnaire [17]. Next, the four hugs are performed. First, the short-duration hug in which the participant activates the closing of the arms with the touch sensor, and after one second, the arms open. Second, the long-duration hug in which the participant activates the closing of the arms with the touch sensor, and after 5 seconds, the arms open. Third, the touch sensor-controlled hug, where the participant controls the opening and closing of the robot's arms using the touch sensor. Finally, the air pressure controlled hug, where the participant must exert force on the balloon located on the robot's back in order to increase the air pressure (above 2.8 psi), and thus control the opening and closing of the arms.

After each of these hugs, the form with the hug perception questionnaire is filled in. At the end of the four hugs, the perception of the social robot survey is filled out again. In addition, a space for comments is given to the participants to talk about the experience with the robot and the specific opinions of the interactions.

4 Results

The Shapiro-Wilk test was applied to determine the normality of the data, which showed that not all the results had a normal distribution. In addition, the Bartlett test was applied, where groups of data with homogeneous and heterogeneous variances were obtained. Therefore, the Mann-Whitney-Wilcoxon statistical test was applied to compare the results of the forms applied in both, the perception of social robotics and the perception of each hug.

In the time analysis, hugs 1 and 2 had a preset duration time, one second and five seconds, respectively. In hug 3, the mean time was $7.83 \pm 4.62s$. On the other hand, hug 4 had a mean time of $7.7 \pm 4.24s$. These mean times are higher than the long-term hug of five seconds.

4.1 Social robot perception before and after the hug interaction

In the perception of social robotics, eight constructs were assessed. Perceived Safety (PS), Perceived Trust (PT), Attitude Towards Technology (ATT), Anxiety (ANX), Social Influence (SI), Social Presence (SP), Perceived Ease of Use (PEOU), and Perceived Utility (PU). This is compared at two time instants (before and after the hugs intervention), as shown in figure 4.

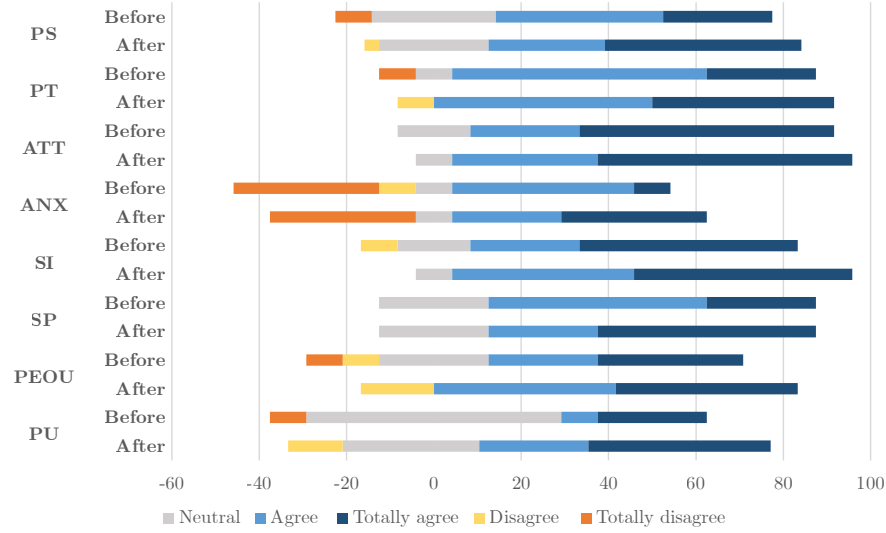


Fig. 4. The survey outcomes of Social Robot perception before and after the hug interaction in the eight constructs evaluated Perceived Safety (PS), Perceived Trust (PT), Attitude Towards Technology (ATT), Anxiety (ANX), Social Influence (SI), Social Presence (SP), Perceived Ease of Use (PEOU) and Perceived Utility (PU).

Considering the p-values obtained, the metrics evaluated had no significant differences in the medians in both moments. Otherwise, PS had a slight increase after having performed the hugging interaction with CASTOR. In the case of ANX, it had an increase after performing the study, which is the opposite of what was expected. In general, there was a positive perception in both instants in the other parameters where the ATT and SP perceptions had no negative opinions, as shown in figure 4.

4.2 Hugs perception

In the perception of hugs, five metrics were evaluated. Friendliness, Safety, Social Intelligence, Enjoyment, and Natural. These metrics were compared between the four hugs implemented, as shown in figure 5.

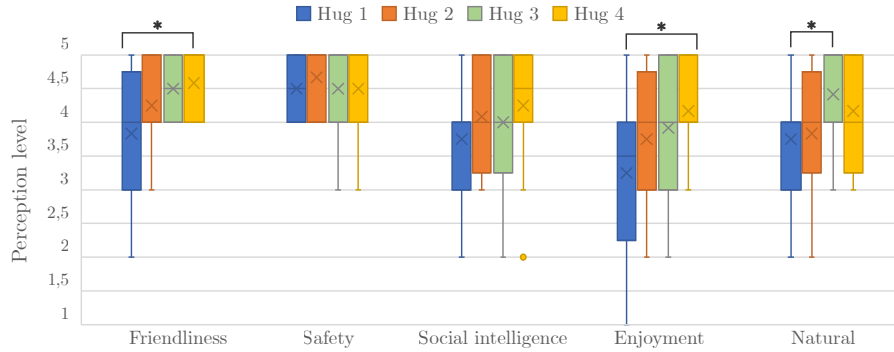


Fig. 5. The survey boxplot after the four hugs, grouped by the five evaluation metrics. The top and bottom of the box represent the 25th and 75th percentile, respectively, while the x mark in the box center represents the mean and the line the median. The lines extending past the boxes show the farthest data point not considered outliers. The points indicate outliers. The * mark means $p < 0.05$

For the Friendliness metric, the comparison between hug 1 and hug 4 had a significant difference ($p=0.036$). This means that hug 4 had a better perception of making the robot seem friendly in the interaction than the preset time hug. In the Safety and Social Intelligence metrics, there were no significant differences, as shown in figure 5. In hug 2, hug 3, and hug 4 medians in perceived safety were precisely the same. In this case, hug 1 had the lowest median. In the case of hug 1, hug 2, and hug 3 medians in perceived social intelligence, the medians were equal. In contrast, hug 4 had the highest perceived social intelligence median, as shown in figure 5.

In the perception of enjoyment, comparing hug 1 and hug 4, significant differences were obtained ($p=0.039$). This represents that the hug 4 was more enjoyable than the hug 1. The lowest median for the enjoyment metric was of hug 1. Finally, concerning the perception of the nature of the hug, the comparison between hug 1 and hug 3 obtained differences ($p=0.047$). This means that hug 3 had a higher perception of natural than hug 1, as seen in figure 5.

5 Discussion

Regarding the duration of the hugs, it could be seen that the hugs controlled by the participants had an even longer time than the long duration time implemented in Hug 2. This is reflected in the results of figure 5, where the perception of the timed release hugs was lower than those controlled by sensors. The average time value greater than 7 seconds in these two controlled-release hugs can be analyzed with the results of friendliness, enjoyment, and natural metrics where participants felt a better interaction when they decided how long to hug the robot.

Block et al. [21] indicate participants would prefer a robot that releases them from a hug immediately when they indicated they were ready for the hug to be over, rather than the robot releasing a hug before or after this time. In this study, we found similar results; the pressure-controlled hug had better perception than a short-time hug in friendliness and enjoyment. In the same way, the touch-controlled hug had a better perception than a short-time hug in nature. Moreover, they indicate they received negative verbal comments during the trials regarding the too short and too long hugs; conversely the investigators did not receive any negative comment about any hug.

In the social robotics perception, in both time instants, positive results were obtained, shown in figure 4, so there were no significant differences. However, in ANX, an increase in the perception of fear of damaging something was obtained after performing the four hugs.

Conclusions and future work

This work presents a contribution by implementing a low-cost social robot in the investigation of physical human-robot interaction through hugs. These interactions were evaluated in four different release methods, two controlled by time and 2 controlled by sensors. From the results obtained, it can be concluded that, as in other studies, the robotic hugs with the best perception are those in which the individual can decide when to end the physical interaction, as in the third and fourth hug. An attempt was made to cover this field of research with CASTOR in a safe way, which was reflected in the perception of safety, being the parameter with the best score in all the hugs. This makes the CASTOR robot a potentially valuable tool in this type of human-robot physical interaction implementing these two controlled release methods.

As part of future work, we want to implement verbal social interaction to improve the perception of the hugs. In addition, this study opens the road to implement CASTOR in a hug interaction study with a group of interest in the emotional and physical needs like children or older population to evaluate the perception of the hugs with the same constructs evaluated in this study.

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